

Jharkhand University of Technology, Ranchi
Diploma 4th Semester Examination, 2024**Subject : Data Base Management System****Subject Code : CSE 404****Time Allowed : 3 Hours****Full Marks : 80****Pass Marks : 26***Answer in your own words.**Answer five questions in which Question No. 1 is compulsory
and answer any four from rest questions.**All questions carry equal marks.***1. Choose the correct option:****2×8=16****(i) What is DBMS?**

(a) DBMS is a collection of queries.

(b) DBMS is a high level language.

(c) DBMS is a programming language.

(d) DBMS stores, modifies and retrieves data.

(ii) In which of the following formats data is stored in the database management system?

(a) Image

(b) Text

(c) Table

(d) Graph

(iii) Which of the following is not a function of the database?

(a) Managing stored data

(b) Manipulating data

(c) Security of stored data

(d) Analysing code

(iv) _____ is a set of one or more attributes taken collectively to uniquely identify a record.

(a) Primary Key

(b) Foreign Key

(c) Super Key

(d) Candidate Key

(v) Which forms have a relation that contains information about a single entity?

(a) 4 NF

(b) 3 NF

(c) 2 NF

(d) 5 NF

(vi) Which of the following is the full form of TCL?

(a) Ternary Control Language

(b) Transaction Control Language

(c) Transmission Control Language

(d) Transaction Central Language

(vii) Rectangles in ER diagrams represent

(a) Tables

(b) Entity sets

(c) Attributes

(d) Tuples

(viii) To remove a relation from an SQL database we use the _____ command.

(a) delete

(b) purge

(c) remove

(d) drop table

2. (a) What is a database system? Give any four features that a database system should provide to its users. 4
- (b) Differentiate between domain, attribute, tuple and relation. 4
- (c) What is a deadlock? State the necessary conditions for deadlock to happen. 4
- (d) Differentiate between schema and instance. 4
3. (a) Explain about the different normal forms in DBMS with suitable examples. 8
- (b) Explain functional dependency and explain its types. Discuss the properties of functional dependency. 4
- (c) Discuss the importance of avoiding NULL values in database. 2
- (d) Explain the role of database administrator. 2
4. (a) Write in detail about the different types of constraints that can be specified on a relation. 8
- (b) What is the difference between DELETE, TRUNCATE and DROP statement in SQL? 3
- (c) Discuss about the different SQL data types with example. 5
5. (a) Explain data fragmentation, data replication and data allocation with its advantages and disadvantages in distributed database. 8
- (b) Write short notes on: 2×4=8
 - (i) Data Manipulation Language
 - (ii) Data Definition Language
 - (iii) Transaction Control Language
 - (iv) ACID properties
6. (a) Explain in detail about the three-tier schema architecture of DBMS. 8
- (b) Explain the client server database architecture in DBMS with its working. 8